

Startup Investment Analysis

Shark Tank Investment Trends — Python, Matplotlib & Tableau

Project Objective

The objective of this project is to analyze startup investment trends, deal success, industry performance, Shark participation, and the relationship between founder asking amounts and actual deal amounts using a Shark Tank startup dataset.

Tools Used

Python, Pandas, Matplotlib, and Tableau.

1. Dataset Overview

The dataset contains startup-level information including startup/brand names, industry, deal status, deal amount, asking amount, and selected Shark participation fields. The analysis was performed across 117 startup records.

2. Key Performance Indicators

Metric	Result
Total startups	117
Startups receiving a deal	65
Deal success rate	55.6%
Total recorded deal amount	3,742.00106

3. Industry-wise Investment Trends

Industry analysis was performed using total deal amount and deal counts. Technology, Food & Beverage, Transportation, Beauty & Personal Care, and Travel & Tourism were among the industry categories examined. The Tableau dashboard provides the visual comparison of investment levels across industries.

Interpretation: Industry-level comparison helps identify where Shark Tank investment capital is concentrated and which domains attract more investor interest.

4. Deal Success Analysis

The Deal field was treated as a binary indicator where 1 represents a successful deal and 0 represents no deal. With 65 successful deals out of 117 startups, the calculated deal success rate is approximately 55.6%.

5. Top Startup Investment Analysis

A Top 10 analysis was created in Tableau using Brand Name and Deal Amount. The visualization ranks startups according to their recorded investment amounts, allowing the largest individual investments to be identified quickly.

6. Average Investment by Industry

Average Deal Amount was calculated for each industry. This differs from total investment: total investment measures the overall capital allocated to an industry, while average investment indicates the typical investment size among records in that industry.

7. Ask Amount vs Deal Amount

A scatter plot was created with Pitcher Ask Amount on the X-axis and Deal Amount on the Y-axis. The observed pattern in the Tableau visualization trends upward, suggesting a positive relationship: startups requesting larger amounts generally tend to receive larger deal amounts. This is an observed association and should not be interpreted as proof that a larger ask directly causes a larger investment.

8. Shark Participation

The available Shark participation fields were analyzed for Piyush, Namita, and Ghazal. The results were Piyush: 27 deals, Namita: 22 deals, and Ghazal: 7 deals. Among these three Sharks, Piyush had the highest recorded deal participation.

Shark	Number of deals
Piyush	27
Namita	22
Ghazal	7

9. Founder Success Pattern Summary

The project focuses on identifying patterns associated with startup funding success. The strongest directly measurable success indicator in the available data is whether a startup received a deal. Asking amount is also useful for understanding funding outcomes. A higher asking amount is associated with higher deal amounts in the scatter plot, while deal status provides the clearest binary success measure.

Important limitation: A detailed founder-profile analysis requires reliable founder attributes such as founder count, age, experience, education, gender, or prior startup history. These should not be inferred when the corresponding fields are absent or incomplete.

10. Key Findings

- 117 startups were included in the analysis.
- 65 startups received deals, giving a deal success rate of approximately 55.6%.
- Total recorded deal amount was 3,742.00106 in the dataset's amount units.
- The Tableau dashboard compares investment concentration across industries.
- The Top 10 visualization identifies startups with the highest recorded investment.
- The Ask Amount vs Deal Amount scatter plot shows an upward/positive pattern.
- Among Piyush, Namita, and Ghazal, Piyush had the highest recorded deal participation.

11. Conclusion

The analysis demonstrates how startup investment data can be transformed into actionable business insights using Python and Tableau. Python and Matplotlib were used for data preparation and exploratory visualization, while Tableau was used to build an interactive dashboard. The results highlight overall deal

success, investment concentration, top investments, Shark participation, and the relationship between asking and received amounts.

12. Tableau Dashboard

The final Tableau dashboard combines the key KPIs and visual analyses into one view. For dashboard design, the project prioritizes the most important metrics at the top and uses charts according to the analytical question. Tableau recommends keeping dashboards focused, using appropriate chart types, and limiting unnecessary views for clarity.

13. Recommended Submission Files

1. Python analysis file (.py or .ipynb)
2. Tableau packaged workbook (.twbx)
3. PDF project report
4. Original/cleaned dataset (.csv or .xlsx)
5. README.md with project description and tools used

References

Tableau Help — Best Practices for Effective Dashboards and Visual Best Practices. These resources emphasize choosing visualizations based on the analytical question, clear layouts, and limiting unnecessary views.